Missing Coin Sum Queries

Time limit: 1.00 s **Memory limit:** 512 MB

You have n coins with positive integer values. The coins are numbered $1, 2, \dots, n$.

Your task is to process q queries of the form: "if you can use coins $a \dots b$, what is the smallest sum you cannot produce?"

Input

The first input line has two integers n and q: the number of coins and queries.

The second line has n integers $x_1, x_2, \dots, x_n$: the value of each coin.

Finally, there are q lines that describe the queries. Each line has two values a and b: you can use coins $a \dots b$.

Output

Print the answer for each query.

Constraints

- $1 \le n, q \le 2 \cdot 10^5$
- $1 \le x_i \le 10^9$
- $1 \le a \le b \le n$

Example

Input	Output
5 3 2 9 1 2 7 2 4 4 4 1 5	4 1 6

Explanation: First you can use coins $[9, 1, 2]$, then coins $[2]$ and finally coins $[2, 9, 1, 2, 7]$.